PROBLEM 11

Conservative Force and Potential II

Original statement · Problems.pdf, p. 9

1D Energy Landscapes

The potential energy function of a particle of mass m is:

$$V(x) = -\frac{c}{2}(x^2 - a^2)^2$$

where c and a are positive constants. Sketch this function, and describe the possible types of motion in the three cases:

(a) $E > 0$

(b) $E < -\frac{ca^4}{2}$

(c) $-\frac{ca^4}{2} < E < 0$

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** a particle of mass m on the Ox axis with the potential energy $V(x) = -\dfrac{c}{2}\left(x^2 - a^2\right)^2, c > 0, a > 0$.
- **Constraints:** motion along the axis ($y = z = 0$, smooth guide); $\mathrm{CDOF} = 1$, coordinate x.
- **Hypotheses:** the force derives from V, so the energy E is a constant; real motion needs $T = \frac{1}{2}m\dot{x}^2 \geq 0$, i.e. $V(x) \leq E$.
- **Parameters:** c [J/m^4], a [m], E [J].

PHASE 2

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{F}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ with $\mathbf{i}$ along the axis (constant unit vectors).

PHASE 3

Derivation

3.1 Kinematics

$$\mathbf{r} = x\,\mathbf{i}, \qquad \mathbf{v} = \dot{x}\,\mathbf{i}, \qquad \mathbf{a} = \ddot{x}\,\mathbf{i}$$

3.2 Forces

- **Applied force:** $F(x) = -\dfrac{dV}{dx}$ (conservative). **Constraint forces:** the reaction of the guide, perpendicular to Ox. **Inertia forces:** none.
- **Force from the potential.** The chain rule on $\left(x^2 - a^2\right)^2$ gives

$$\frac{dV}{dx} = -\frac{c}{2} \cdot 2\left(x^2 - a^2\right) \cdot 2x = -2cx\left(x^2 - a^2\right) \implies F(x) = 2cx\left(x^2 - a^2\right)$$

3.3 Equations of Motion

- **Newton's second law:** $m\ddot{x} = F(x) = -V'(x)$. Multiplying by $\dot{x}$ and using the chain rule, $\frac{d}{dt}\left(\frac{1}{2}m\dot{x}^2 + V(x)\right) = \dot{x}\left(m\ddot{x} + V'(x)\right) = 0$:

$$E = \tfrac{1}{2}m\dot{x}^2 + V(x) = \text{const}, \qquad \text{so} \quad T(x) = E - V(x) \geq 0$$

- The turning points ($T = 0$) are the roots of $V(x) = E$, and the allowed regions are those with $V(x) \leq E$.

3.4 Integration and Answers

Equilibria and sketch of the potential

> **Pedagogical Justification:** the equilibrium points are where the force vanishes, $dV/dx = 0$, and their stability follows from the sign of V'' (minimum: stable; maximum: unstable):

$$\frac{dV}{dx} = 0 \iff x = 0,\ \pm a, \qquad \frac{d^2V}{dx^2} = \frac{d}{dx}\left(-2cx^3 + 2ca^2x\right) = -6cx^2 + 2ca^2 = 2c\left(a^2 - 3x^2\right)$$

- **At $x = 0$:** $V(0) = -\frac{1}{2}ca^4$ and $V''(0) = 2ca^2 > 0$: a local minimum (stable equilibrium).
- **At $x = \pm a$:** $V(\pm a) = 0$ and $V''(\pm a) = 2c\left(a^2 - 3a^2\right) = -4ca^2 < 0$: local maxima (unstable equilibria, potential barriers).
- **Symmetry and limits:** $V(-x) = V(x)$, and $V \to -\infty$ as $x \to \pm\infty$. The sketch is built from these values: a central well and two barriers, with $V \to -\infty$ on both sides.

x	$-\infty$	$-a$	0	$+a$	$+\infty$
$V(x)$	$-\infty$	0 (maximum)	$-\frac{1}{2}ca^4$ (minimum)	0 (maximum)	$-\infty$

The curve is even: it falls from the left maximum at $-a$ to the central minimum and rises again to the maximum at $+a$, forming a well of depth $\frac{1}{2}ca^4$ between two barriers of height 0, and then falls without bound.

Case (a): $E > 0$

- The maximum of the potential is $V_{\max} = 0 < E$, so $E > V(x)$ for every x and $V(x) = E$ has no real root (since $-\frac{c}{2}(x^2 - a^2)^2 \leq 0 < E$). Then

$$T(x) = E - V(x) = E + \frac{c}{2}\left(x^2 - a^2\right)^2 \geq E > 0$$

(equality only at $x = \pm a$), so the velocity never vanishes or changes sign.

- **Type of motion: unbounded.** The particle moves toward $+\infty$ or toward $-\infty$ according to the sign of its velocity. A particle moving toward the origin crosses the central region and both barriers with $v \geq \sqrt{2E/m}$.

Case (b): $E < -\frac{1}{2}ca^4$

- The energy is below the bottom of the well, $E < V(0)$. For $|x| \leq a$ the potential satisfies $V \geq V(0) > E$, so $T < 0$ there: the central region is forbidden.
- **Turning points.** $V(x) = E$ means $\left(x^2 - a^2\right)^2 = -\dfrac{2E}{c} = \dfrac{2|E|}{c}$. Taking the square root, $x^2 - a^2 = \pm\sqrt{2|E|/c}$. The root with the minus sign gives $x^2 = a^2 - \sqrt{2|E|/c} < 0$ (because $|E| > \frac{1}{2}ca^4$ implies $\sqrt{2|E|/c} > a^2$), which has no real x, so only the plus sign remains:

$$x^2 = a^2 + \sqrt{\frac{2|E|}{c}} \implies x = \pm x_{\text{out}}, \qquad x_{\text{out}} = \sqrt{a^2 + \sqrt{\frac{2|E|}{c}}} > a$$

- **Allowed regions:** $x \leq -x_{\text{out}}$ and $x \geq +x_{\text{out}}$.
- **Type of motion: unbounded.** A particle in the right region that moves toward $+\infty$ simply goes to $+\infty$. If it moves toward the origin it decelerates, stops at $+x_{\text{out}}$ and is pushed back by the force $F = -V'(x_{\text{out}}) > 0$ to $+\infty$ (reflection). The left region is symmetric.

Case (c): $-\frac{1}{2}ca^4 < E < 0$

- The energy lies between the well bottom and the barrier tops: $V(0) < E < V(\pm a)$. The roots of $V = E$ are

$$x^2 = a^2 \pm \sqrt{\frac{2|E|}{c}}, \qquad \sqrt{\frac{2|E|}{c}} < a^2 \ \text{ because } |E| < \tfrac{1}{2}ca^4$$

so both values of x^2 are positive: four turning points $\pm x_{\text{in}}$ and $\pm x_{\text{out}}$, with $x_{\text{in}} = \sqrt{a^2 - \sqrt{2|E|/c}} \in (0, a)$ and $x_{\text{out}} = \sqrt{a^2 + \sqrt{2|E|/c}} > a$, ordered $-x_{\text{out}} < -x_{\text{in}} < 0 < x_{\text{in}} < x_{\text{out}}$.

- **Allowed regions** ($V \le E$): the inner well $[-x_{\text{in}}, +x_{\text{in}}]$ and the two outer regions $|x| \ge x_{\text{out}}$.
- **Types of motion.** In the well, the particle is confined and **oscillates** periodically between $-x_{\text{in}}$ and $+x_{\text{in}}$ around the stable equilibrium $x = 0$. In an outer region it behaves as in case (b): it moves directly toward $\pm\infty$ or is reflected at $\pm x_{\text{out}}$ back to $\pm\infty$.

PHASE 4

Interpretation & Checks

- **Summary:**

Energy range	Turning points	Allowed regions	Motion
(a) $E > 0$	none	all of $\mathbb{R}$	unbounded, goes to $\pm\infty$
(b) $E < -\frac{1}{2}ca^4$	$\pm x_{\text{out}}$	$x \le -x_{\text{out}}$ or $x \ge x_{\text{out}}$	unbounded, directly or reflected
(c) $-\frac{1}{2}ca^4 < E < 0$	$\pm x_{\text{in}}, \pm x_{\text{out}}$	$[-x_{\text{in}}, x_{\text{in}}]$ or $\lvert x\rvert \ge x_{\text{out}}$	oscillation in the well, or unbounded (directly or reflected)

- **Limiting energies:** for $E \to -\frac{1}{2}ca^4$ (bottom of the well), $x_{\text{in}} \to 0$ and $x_{\text{out}}^2 \to 2a^2$: the well collapses to the stable point, whose small oscillations have $\omega^2 = V''(0)/m = 2ca^2/m$. For $E \to 0^-$, $x_{\text{in}}, x_{\text{out}} \to a$: the turning points merge at the unstable equilibria $\pm a$ (separatrix between (c) and (a)).
- **Dimensional check:** $[ca^4] = (\mathrm{J/m^4})\,\mathrm{m}^4 = \mathrm{J}$, like E; $[2|E|/c] = \mathrm{m}^4$, so $\sqrt{2|E|/c}$ is an area and x_{out} is a length.
- **Check against the official key:** the key of `Problems.pdf` is qualitative: (a) the particle moves toward $+\infty$ or $-\infty$; (b) toward $\pm\infty$ directly or after being reflected; (c) toward $\pm\infty$ directly, reflected, or oscillating. This agrees with the classification above. Sympy confirms V', $V''(0)$, $V''(\pm a)$ and the turning points $x^2 = a^2 \pm \sqrt{2|E|/c}$.

Source: Problems.pdf